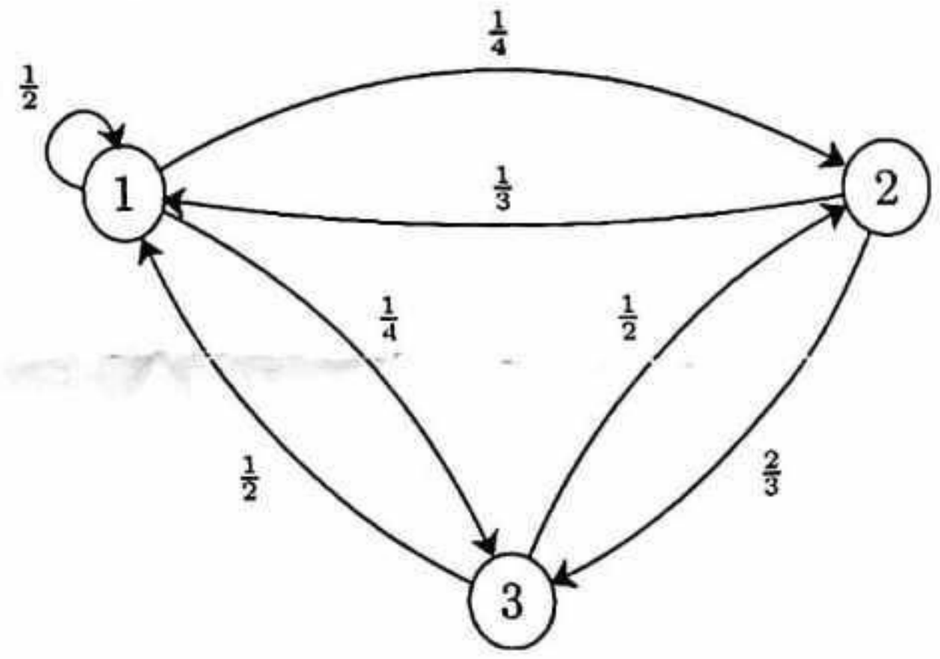
Course Code: CSE-3201(Mathematical Analysis for Computer Science) **Time: 1 Hrs**

1. Three tanks fight a three-way duel. Tank A has probability ½ of destroying the tank at which it fires, tank B has probability 1/3 of destroying the tank at which it fires, and tank C has probability 1/6 of destroying the tank at which it fires. The tanks fire together and each tank fires at the strongest opponent not yet destroyed. Form a Markov chain along with transition matrix by taking as states the subsets of the set of tanks. Hint: Take as states ABC, AC, BC, A, B, C, and none, indicating the tanks that could survive starting in state ABC. You can omit AB because this state cannot be reached from ABC.
2. Find out the transition matrix for the following scenario
   a) The President of the United States tells person A his or her intention to run or not to run in the next election. Then A relays the news to B, who in turn relays the message to C, and so forth, always to some new person. We assume that there is a probability that a person will change the answer from yes to no when transmitting it to the next person and a probability b that he or she will change it from no to yes. We choose as states the message, either yes or no.
   b) In the Dark Ages, Harvard, Dartmouth, and Yale admitted only male students. Assume that, at that time, 80 percent of the sons of Harvard men went to Harvard and the rest went to Yale, 40 percent of the sons of Yale men went to Yale, and the rest split evenly between Harvard and Dartmouth; and of the sons of Dartmouth men, 70 percent went to Dartmouth, 20 percent to Harvard, and 10 percent to Yale.
   c) Write a program to compute $u^{(n)}$ given u and P.



3. Consider the following markov chain
   a) Is this chain irreducible?
   b) Is this chain aperiodic?
   c) Find the stationary distribution for this chain.
   d) Is the stationary distribution a limiting distribution for the chain?
4. Define Queuing network. Write short notes on different customer behavior in queuing model.
5. The arrival rate of customers is 2 per minute and the service rate of each server is 3 per minute. Find the following:
   a) The average number of customers in the system.
   b) The average waiting time in the queue.
   c) The system utilization.
   d) The probability of a customer finding the server busy.
   e) The probability of customers waiting in the queue.
   f) The probability of customers being blocked.
6. A company has positions for three employees. Applicants for these jobs appear at a Poisson rate of two per year. If all jobs are filled, applicants look elsewhere. Those who hold a job do so for an exponentially distributed time with mean 1 / 2 year.
   a) Describe the situation, using Kendall's notation, as a queueing system.
7. An airport has two runways, both of which can be used for both take offs and landings. Each plane, whether taking off or landing, occupies a runway for an average of two minutes, where occupy means "prevent use by other planes". Delay on the ground, although unpleasant for passengers, is not a safety hazard; delay in the air, on the other hand, is of serious concern. Suppose that F.A.A. regulations specify that the mean delay in the air (i.e., Wq) must not exceed ten minutes. Assume that planes arrive according to a Poisson process with rate 20 planes per hour. We are interested in an M/G/c /N model to study the maximum load in terms of the mean number of planes that arrive per hour; that is, such that Wq= 10 minutes.
   a) What is your value of c?
   b) What is your value of N?
   c) What is your value of your service rate, μ?

Scanned with CamScanner
AVAILABLE AT: Onebyzero Edu - Organized Learning, Smooth Career (www.onebyzeroedu.com)